Figure 1. One platform, two cooperating modes, in a live energy OT environment

SOC Mode = deception-based active defence. Research Mode = offline analyst copilot. They share one data layer: live attacker behaviour informs research, research insight informs defence.

Energy operator OT environment (Purdue model)

Enterprise / IT (L4-L5)
Business network, historian access

OTShield decoys (protocol-aware)

DMZ / Supervisory (L3)
SCADA, PI Historian, eng. workstations

OTShield decoys (protocol-aware)

Control (L1-L2)
HMIs, PLCs, RTUs - Modbus, S7, DNP3, OPC UA, IEC 104

OTShield decoys (protocol-aware)

SOC Mode

DECEPTION-BASED ACTIVE DEFENCE

- Protocol-aware decoy fabric - any contact is hostile
- Purdue-aware deep-packet inspection + asset inventory
- MITRE ATT&CK for ICS mapping; incident triage
- High-confidence, low-noise alerts to SIEM / SOAR

shared data layer

Research Mode

OFFLINE ANALYST COPILOT

- Workbench: Library, Summary, Inventory, Threads, Findings, Vulns
- Local LLM summarises evidence; answers with citations
- Turns attacker activity into auditable Findings
- Safe threat hunting; surfaces legacy-system exposure

Operator SOC + analyst team -> proactive defence, incident response and safe threat hunting
Coexists with Dragos / Claroty. No risk to live controllers. IOC export to SIEM / SOAR.

Air-gap-safe threat-intel feed
OTShield's live internet-exposed decoy fabric captures real ICS attacker tactics, packaged as signed update bundles ingested offline.

Table 1. Mapping to the four competition challenges

Challenge	Mode	How OTShield delivers it in this project
1. Proactive CNI defence & incident response	**SOC Mode**	High-confidence decoy detections + DPI; alerts to SIEM/SOAR; triage and response with the operator SOC team.
2. Detection & eviction of sophisticated actors	**SOC Mode**	Surfaces protocol-aware intent (unauthorised writes, interlock overrides, remote discovery) mapped across ATT&CK for ICS.
3. Threat hunting for CNI organisations	**Research Mode**	Copilot lets analysts query detections, manuals and asset data together and follow leads safely, with zero risk to live controllers.
4. Legacy system vulnerabilities in CNI	**Both modes**	SOC Mode inventories and risk-scores legacy assets; Research Mode copilot reasons over them to show how they could be reached and abused.

Figure 2. Readiness: TRL 7 today -> TRL 8 at project end

Today (TRL 7)
Both modes operational: decoy fabric validated against real ICS attackers (157 / 20 countries / 12,968 attacks); Research Mode copilot working offline.

Q1-Q2 (TRL 7)
Deploy both modes into the energy operator OT environment; tune to real assets, protocols and legacy equipment; first detection and hunting campaigns.

Q3 (TRL 7-8)
Continuous operation; incident-response and hunting exercises; capture detection-rate, time-to-detect and analyst-time-saved metrics; security review.

Q4 (TRL 8)
Qualified in operational environment; operator acceptance; commercialisation plan and development case study delivered.

Why it meets the challenge
One integrated platform - SOC Mode for deception-based active defence, Research Mode for the analyst copilot - is proven, accepted and qualified inside a real UK energy operator, delivering all four challenge areas and ready to scale across the sector.